



Development and multi-institutional validation of the Swin transformer model for prediction of spontaneous passage of ureteral calculi: A retrospective multicentric study

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Abstract

To improve the accuracy of our prediction of the spontaneous passage of urinary calculi, we developed a model based on the Swin Transformer model. In this retrospective multicentre study, computed tomography (CT) images of ureteral calculi from January 2019 to December 2023 from our hospital were used as internal training and validation datasets. The samples were split in a 7:3 ratio into an internal training set ($N=278$) and internal testing set ($N=120$). An external validation set ($N=142$) was created using the CT datasets of patients from four different hospitals. Swin Transformer was used to create an ensemble learning-based model approach for predicting the passage of ureteral calculi. For both the internal and external validation sets, performance was assessed using the F1 score, accuracy, specificity, sensitivity, and area under the operating characteristic curve (AUC). A total of 540 urinary CT images were acquired (Internal dataset=398; multicentre external dataset=142). The AUC reached 0.873 (95% [CI]: 0.841–0.906), and the external validation set reached 0.784 (95% CI: 0.747–0.820). The evaluation results of the Swin Transformer model, developed using CT images of urinary calculi, demonstrate that it outperforms models such as ResNet and ConvNeXt. Furthermore, the model outperforms both attending and resident doctors. The predictive power of attending doctors was significantly improved with the aid of the Swin Transformer model. The Swin Transformer model is a generalizable, objective and accurate prediction model for calculi passage based on urinary tract CT images and could help clinicians make better clinical judgements.

Trial registration: It was also registered in the Chinese Clinical Trial Registration (registration number: ChiCTR2400086807). Informed consent was waived as this study used retrospectively collected anonymized data.

Keywords Ureteral calculi · Deep learning model · Computed tomography · Spontaneous passage

Abbreviations

AI	Artificial intelligence	AR	Attending radiologist
DL	Deep learning	SR	Senior radiologist
AUC	Area under the curve	GRAD-CAM	Gradient-weighted class activation mapping .
CT	Computed tomography		
ROC	Receiver operating characteristic curve		
RU	Resident urologist		
AU	Attending urologist		
SU	Senior urologist		
RR	Resident radiologist		

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Introduction

Ureteral calculi are a common urological condition, and determining the optimal treatment strategy (conservative management versus surgical intervention) remains a significant challenge [1]. The likelihood of spontaneous passage of ureteral calculi is influenced by a complex interplay of factors, including stone size, location, composition, and anatomical considerations [2, 3]. Although medical research has made significant progress in understanding the factors influencing ureteral stone passage, the lack of standardized and effective predictive tools remains a major limitation. Currently, clinicians primarily utilize the size and location of ureteral calculi as key criteria for determining the likelihood of spontaneous passage, which in turn informs the selection between conservative management and surgical intervention. Nevertheless, this approach is associated with notable limitations. In some cases, patients may be inappropriately assigned to conservative treatment due to inaccurate assessments, leading to ineffective stone expulsion and an increased risk of severe complications, including acute renal failure, obstructive pyelonephritis, and sepsis. Conversely, other patients may undergo surgical procedures prematurely, resulting in unnecessary healthcare expenditures and the potential for overtreatment.

Recent advances in artificial intelligence (AI) have revolutionized medical image analysis, leading to the development of innovative diagnostic and prognostic models [4]. Deep learning analysis of images enables the extraction of disease-related information, which can be used to improve the objectivity and accuracy of diagnosis, reduce the time and costs involved in diagnosis, and help prescribe correct treatments [5–7]. Deep learning models have demonstrated significant potential in assessing the efficacy of various treatment modalities for ureteral calculi, including extracorporeal shock wave lithotripsy, ureteroscopy, and ureterolithotripsy. However, these models have typically been constrained by limited sample sizes and data sourced from a single medical center, resulting in relatively poor generalization performance [2, 4, 8–10]. This study proposes a novel approach for predicting the spontaneous passage of ureteral calculi using the Swin Transformer, a deep learning architecture capable of capturing long-range dependencies in visual data. Our model leverages the Swin Transformer to extract high-level representations from both axial and coronal CT images of the same patient, effectively modeling the relationships between these representations. The model's discriminative power, calibration capacity, and clinical utility are enhanced through external validation.

We evaluated our model on an internal dataset and observed promising results in predicting calculi passage. Furthermore, external validation with four collaborating

medical centers demonstrated the robustness and generalizability of our approach. Artificial intelligence assisted clinicians' study further revealed that our model has the potential to improve inter-observer agreement among clinicians.

Materials and methods

Datasets

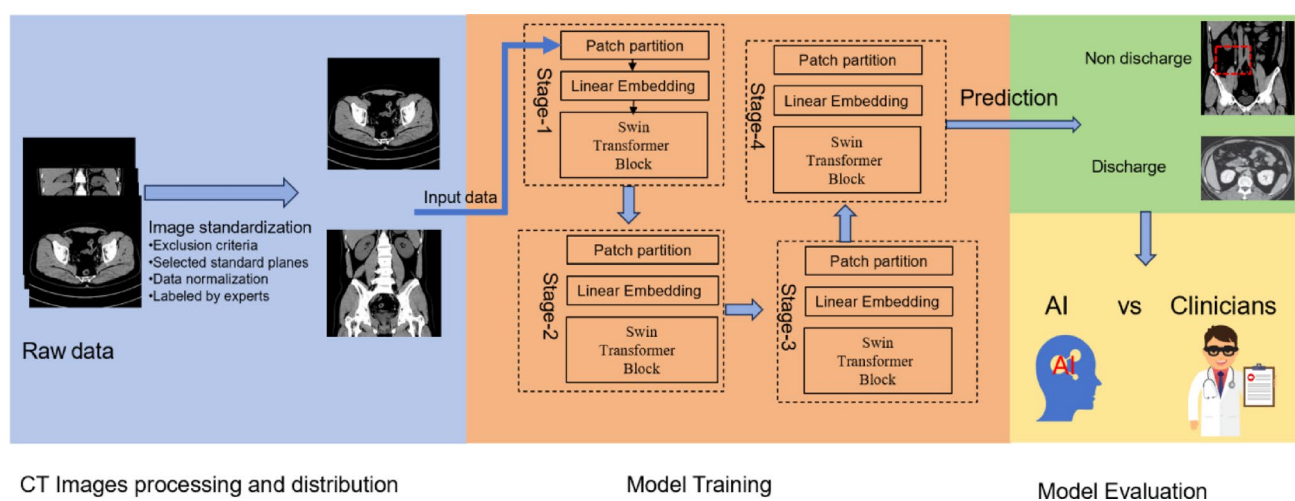
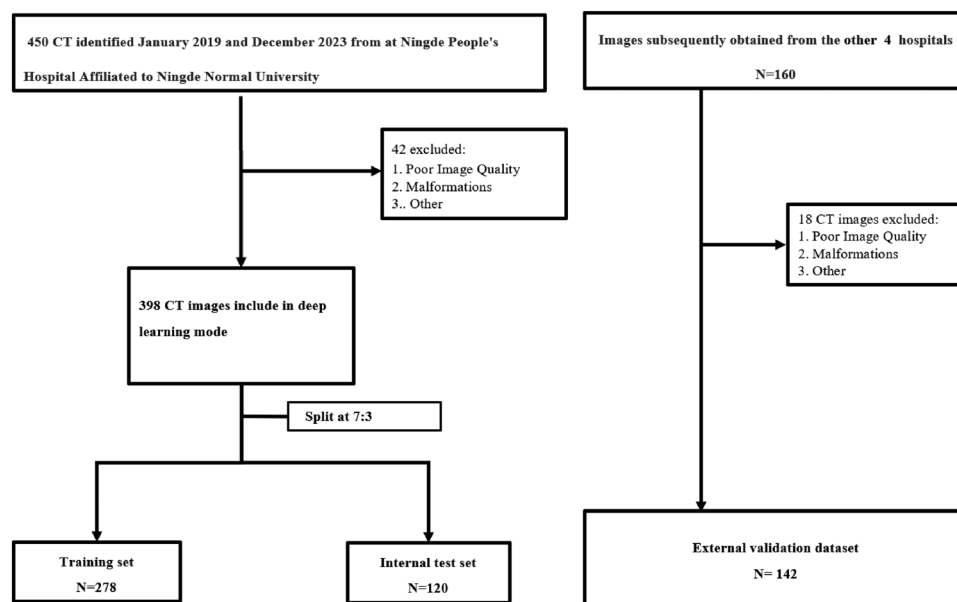
From January 2019 to December 2023, a total of 540 CT image datasets were retrospectively collected from 5 hospitals as part of a retrospective observational study. The inclusion and exclusion criteria applied for participant selection in the datasets are detailed in the Appendix (Table 4). We used CT datasets from 4 independent medical institutions as external test cohorts to assess the performance of the model: Pingnan County General Hospital (Fujian, China), XiaPu County General Hospital (Fujian, China), The First Affiliated Hospital of Hebei North University (Hebei, China), and Guizhou Aerospace Hospital (Guizhou, China); the data are described in the Supplementary material (Table 1).

Image annotation and labeling

All included cases that met the inclusion criteria were consecutive. For each patient, the non-contrast CT image was labeled according to the stone expulsion status confirmed by one month follow-up imaging. A binary label was assigned: 1 (stone expelled) or 0 (stone retained). Only images with clear visualization of the ureteral stone region, free from motion artifacts or interfering devices (e.g., DJ stents, nephrostomy tubes), were included. To ensure consistency and image quality, two representative axial slices per patient were selected by two board-certified radiologists with more than 10 years of experience in urogenital imaging. These slices were selected using standardized methods. For each case, one coronal image showing the maximum diameter of the ureteral stone along the long axis and one axial image showing the maximum diameter of the cross-section were chosen. Cases with ambiguous findings or missing follow-up were excluded from the dataset.

Data processing

To enhance generalization and reduce overfitting, we applied a series of data augmentations. Input images were randomly resized and cropped to 224×224 pixels (scale: 0.5–2.0, bicubic interpolation), followed by random horizontal and vertical flips (50% each). Additional transformations included

Table 1 Flowchart of participant enrolment and selection in this study from the CT database**Fig. 1** The architecture of the model

Gaussian blur (20% chance, 5×5 kernel, $\sigma=0.1-3.0$), and $\pm 15^\circ$ rotation. Images were then converted to tensors and normalized. All augmentations were applied sequentially using composed transforms.

Deep learning

Model architecture

The Swin Transformer, a model tailored for computer vision, excels in classification tasks by using a shifted window self-attention mechanism that captures both global context and local features. This capability makes it particularly effective

for handling multiscale features in high-resolution images, thereby improving classification accuracy.

In this study, we use the Swin Transformer to analyse axial and sagittal CT images from each patient to predict the likelihood of stone passage. Each image is processed independently by separate Swin Transformer networks, generating 1000-dimensional feature vectors. These vectors are then integrated using a MLP module, which computes attention weights to emphasize the most informative elements. The weighted feature vectors are fused into a single vector, which is used for classification and analysis.(Fig. 1).

Training settings

The training objective is to match the categorical output to the smoothed labels. We use a batch size of 16 and train the model for a total of 50 epochs. The initial 10 epochs are dedicated to learning rate warm-up, starting at 0 and gradually increasing to $5e-4$. This is followed by a cosine annealing schedule for the remaining 40 epochs, where the learning rate decays from $5e-4$ to $1e-6$. To enhance model generalization and prevent overfitting, we incorporate label smoothing, which softens the ground-truth labels in the training data. This technique regulates the output distribution and makes the model more likely to produce more robust predictions.

Model performance and comparison with clinicians

The classification performance of the proposed model was first evaluated using an internal dataset and compared against several baseline models. To assess the impact of model assistance on clinical decision-making, an external dataset was used in a reader study involving six clinicians. Each clinician independently reviewed the CT images to determine stone expulsion status, both without and subsequently with model assistance. Notably, no washout period was applied between the unaided and aided assessments. During the model-assisted phase, clinicians were provided with the model's predicted probability scores and heatmap visualizations. Based on this information, they could either retain or revise their initial decisions. Comparative analysis between the Swin Transformer model and clinicians was conducted using the area under the receiver operating characteristic curve (AUC), as well as accuracy, sensitivity, and specificity.

Table 2 Demographics of the data sets from the CT image internal and the external cohort

Patient Characteristics	Training Set (<i>N</i> =278)	Internal test set (<i>N</i> =120)	External Test Set (<i>N</i> =142)
Patients	278	120	142
Age (years, mean±SD)	45.05±14.43	43.80±12.71	48.96±11.77
Sex			
Male	223	93	88
Female	55	27	54
Whether the stone passed			
No	111	48	91
Yes	167	72	51

Statistical analysis

Statistical analyses were performed using Python 3.8 (<https://www.python.org/>), primarily with core libraries such as scikit-learn (<https://scikit-learn.org/>), SciPy (<https://www.scipy.org/>), and Matplotlib (<https://matplotlib.org/>). Additional auxiliary software packages were also utilized. Model performance in classification tasks was evaluated using a comprehensive set of metrics: area under the receiver operating characteristic curve (AUC), sensitivity, specificity, F1-score, precision, and accuracy. To obtain stable estimates and quantify uncertainty, 1,000 bootstrap resamples were conducted to derive confidence intervals for each metric. Statistical significance was determined using a two-sided *P* value threshold of 0.05.

Results

Patient characteristics

A total of 540 CT images were collected throughout the duration of the study. After the eligibility criteria were applied, 398 images were included (mean age: 44.6 ± 12.11 months; 316 males [79.40%], 82 females [20.60%]). Participants were allocated into training (70%, *N*=278) and testing (30%, *N*=120) cohorts using a split-sample validation approach. For the external test set, the mean age was 48.96 ± 11.77 months, comprising 88 males (61.97%) and 54 females (38.03%). Detailed demographic characteristics are summarized in Table 2.

Performance of the Swin transformer model on the internal and external datasets

The confusion matrices generated using the internal and external test data are shown in Fig. 2. In the internal dataset, the model achieved a sensitivity of 0.902 (95% CI: 0.864–0.939), a specificity of 0.705 (95% CI: 0.646–0.759), and an AUC of 0.873 (95% CI: (0.841–0.906)). In the external dataset, the model achieved an AUC of 0.784 (95% CI: 0.747–0.820), a sensitivity of 0.768 (95% CI: 0.722–0.817), and a specificity of 0.658 (95% CI: 0.600–0.714). The results of the model for the four external units are shown in the Appendix section.

Model performance and comparison with clinicians at the external dataset

The clinical prognostic predictions of the six clinician at the external data set, including sensitivity and specificity, are displayed together with the model's ROC curve (Fig. 3). As

Fig. 2 Performance of the model on the internal and external datasets

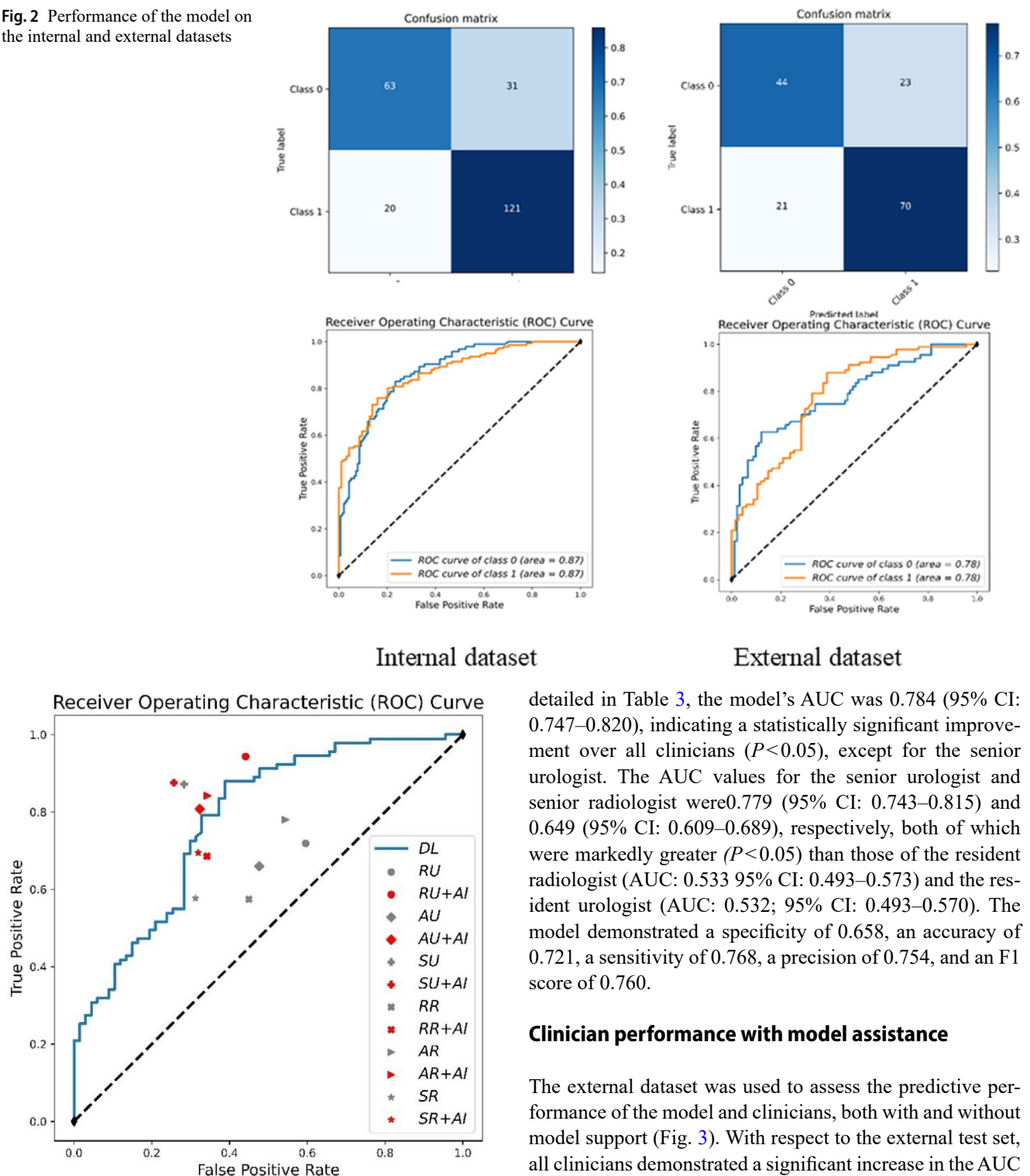


Fig. 3 The clinical prediction performance of six clinicians with and without model assistance on the external dataset. RU, Resident urologist; AU, Attending urologist; SU, Senior urologist; RR, Resident radiologist; AR, Attending radiologist; SR, Senior radiologist; ROC, receiver operating characteristic; AUC, area under receiver operating characteristic curve

detailed in Table 3, the model's AUC was 0.784 (95% CI: 0.747–0.820), indicating a statistically significant improvement over all clinicians ($P < 0.05$), except for the senior urologist. The AUC values for the senior urologist and senior radiologist were 0.779 (95% CI: 0.743–0.815) and 0.649 (95% CI: 0.609–0.689), respectively, both of which were markedly greater ($P < 0.05$) than those of the resident radiologist (AUC: 0.533 95% CI: 0.493–0.573) and the resident urologist (AUC: 0.532; 95% CI: 0.493–0.570). The model demonstrated a specificity of 0.658, an accuracy of 0.721, a sensitivity of 0.768, a precision of 0.754, and an F1 score of 0.760.

Clinician performance with model assistance

The external dataset was used to assess the predictive performance of the model and clinicians, both with and without model support (Fig. 3). With respect to the external test set, all clinicians demonstrated a significant increase in the AUC when aided by the model compared with their unaided performance ($P < 0.05$).

Model visualization

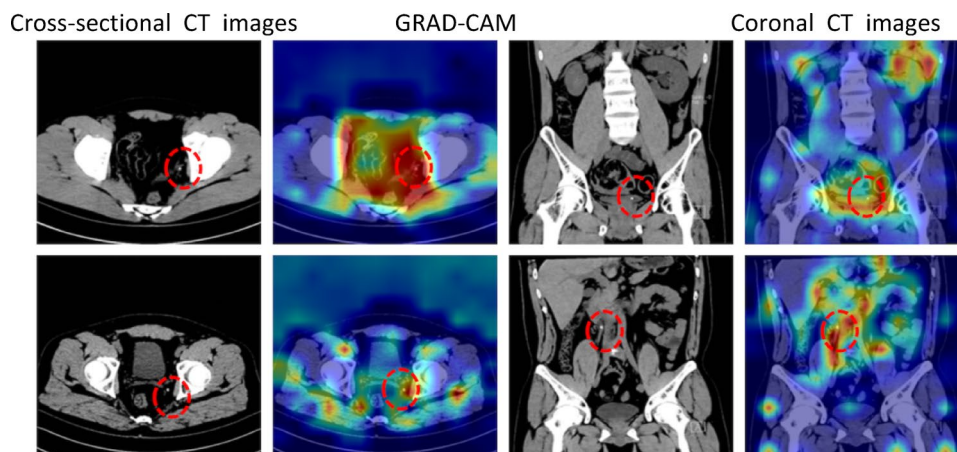
Our model learns directly from the raw or preprocessed image, enabling it to discover latent features that contribute

Table 3 Performance of the Swin transformer model, clinicians alone, and clinicians assisted by the model on the external dataset

	Accuracy (95% CI)	AUROC (95% CI)	F1 Score (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	Precision (95% CI)
DL	0.7214 (0.6835,0.7595)	0.7840 (0.7473,0.8204)	0.7598 (0.7239,0.7957)	0.7683 (0.7216,0.8171)	0.6578 (0.6000,0.7143)	0.7535 (0.7099,0.7976)
RU	0.5450 (0.5070,0.5845)	0.5324* (0.4931,0.5700)	0.6138 (0.5697,0.6550)	0.6643 (0.6104,0.7161)	0.4005 (0.3433,0.4563)	0.5729 (0.5233,0.6222)
AU	0.5279 (0.4859,0.5704)	0.5199* (0.4797,0.5612)	0.5815 (0.5350,0.6265)	0.6035 (0.5466,0.6579)	0.4363 (0.3750,0.5000)	0.5639 (0.5065,0.6220)
SU	0.7862 (0.7535,0.8239)	0.7788 (0.7431,0.8148)	0.8139 (0.7805,0.8471)	0.8571 (0.8182,0.8961)	0.7005 (0.6452,0.7581)	0.7766 (0.7311,0.8242)
RR	0.5358 (0.5000,0.5775)	0.5327* (0.4933,0.5727)	0.5699 (0.5263,0.6164)	0.5635 (0.5060,0.6203)	0.5019 (0.4429,0.5606)	0.5797 (0.5269,0.6351)
AR	0.6119 (0.5704,0.6549)	0.5959* (0.5517,0.6364)	0.6805 (0.6386,0.7210)	0.7554 (0.7032,0.8077)	0.4364 (0.3750,0.5000)	0.6212 (0.5714,0.6703)
SR	0.6476 (0.6056,0.6901)	0.6485* (0.6088,0.6892)	0.6637 (0.6216,0.7067)	0.6388 (0.5854,0.6912)	0.6583 (0.6000,0.7184)	0.6941 (0.6399,0.7500)
RU+DL	0.7946 (0.7606,0.8310)	0.7765 (0.7456,0.8080)	0.8362 (0.8051,0.8667)	0.9604 (0.9389,0.9857)	0.5926 (0.5323,0.6522)	0.7416 (0.6965,0.7858)
AU+DL	0.7596 (0.7254,0.7958)	0.7617* (0.7265,0.7970)	0.7710 (0.7324,0.8098)	0.7419 (0.6923,0.7907)	0.7816 (0.7324,0.8333)	0.8050 (0.7585,0.8529)
SU+DL	0.8366 (0.8028,0.8662)	0.8343* (0.8019,0.8663)	0.8513 (0.8188,0.8834)	0.8577 (0.8193,0.8987)	0.8109 (0.7606,0.8594)	0.8465 (0.8056,0.8874)
RR+DL	0.6987 (0.6620,0.7394)	0.6977* (0.6582,0.7382)	0.7194 (0.6761,0.7615)	0.7075 (0.6500,0.7607)	0.6879 (0.6324,0.7460)	0.7345 (0.6842,0.7838)
AU+DL	0.7618 (0.7254,0.7958)	0.7538* (0.7158,0.7904)	0.7931 (0.7578,0.8276)	0.8338 (0.7901,0.8750)	0.6737 (0.6129,0.7313)	0.7580 (0.7096,0.8065)
SR+DL	0.6985 (0.6620,0.7324)	0.6975* (0.6609,0.7354)	0.7188 (0.6767,0.7598)	0.7048 (0.6533,0.7590)	0.6902 (0.6346,0.7500)	0.7360 (0.6885,0.7850)

The performance differences between clinicians and DL model, as well as those among clinicians, were analyzed, and corresponding P values were calculated. Detailed comparisons are provided in Appendix Tables S2-7. * $P < 0.05$ was considered statistically significant compared with the DL model

Fig. 4 Gradient-weighted class activation mapping, CT image (left) and GRAD-CAM markup (right). The marked position indicates the location of the ureteral calculus



to prediction. To increase clinical interpretability, we highlighted the areas that the model focuses on during prediction by creating heatmaps on the images using gradient-weighted class activation mapping (GRAD-CAM). This graphic displays the areas of the image that have the greatest influence on the model's choice, which may indirectly reflect the model's learned emphasis on factors such as stone size, location, or ureteral dilation (Fig. 4).

Discussion

This research presents an AI model that leverages the Swin Transformer architecture to predict the spontaneous passage of ureteral stones from CT images. Our model, trained and validated on a multicentre dataset, demonstrated promising performance, with AUC of 0.87 and 0.78 on the internal and external test sets, respectively. These results suggest that our model can assist doctors in predicting ureteral calculi

passage and guiding treatment decisions. Currently, clinical guidelines primarily rely on stone size to assess the likelihood of spontaneous passage of ureteral stones. For ureteral stones measuring less than 5 mm in diameter, approximately 75% are expelled spontaneously; therefore, non-surgical management is recommended as the initial treatment approach. For stones ranging from 5 to 10 mm in diameter, non-surgical intervention may also be considered, provided that close clinical monitoring is maintained. Evidence from research suggests that the average time for spontaneous stone passage is approximately 6 to 29 days [11]– [12]. Previous studies exploring the prediction of ureteral calculi passage have relied primarily on regression-based methods and have focused on quantifiable factors such as the size and location of the ureteral calculi, hydronephrosis, and laboratory markers [13–16]. However, these approaches often struggle to incorporate complex information, such as stone shape and its relationship with surrounding tissues. Multi-layer perceptron deep learning exhibited a higher specificity than logistic regression did, particularly for ureteral stones measuring 5–10 mm. Nevertheless, there were no statistically significant differences in the predictive power between this model and logistic regression [2]. While machine learning models incorporating patient demographics and clinical parameters have been proposed, they may not capture the intricate details present in medical images. Xiao et al. [17] used the ResNET model to predict urinary calculi on images, which could accurately determine the prognosis of conservative treatment in patients with ureteral calculi, he anatomical structure of the ureter and the status of hydronephrosis were not incorporated into the considerations. and it is only a single-center clinical study and has not been widely verified.

Our study addresses these limitations by leveraging the Swin Transformer's ability to extract high-level representations from CT images, effectively capturing both local and global contextual information. First, This model requires CT scans of the urinary system to generate a preliminary prediction. The data collection process is straightforward, making it feasible for implementation across medical institutions at all levels and demonstrating strong generalizability. By capturing spatial relationships within the imaging data, our approach goes beyond traditional scoring methods, which are typically limited to linear measurements and empirical formulas. Compared with other classification models (e.g., ResNet and ConvNeXt), the Swin Transformer demonstrated superior performance. Furthermore, the accuracy of the model was increased through multicentre validation across different geographic regions. Second, the model was externally validated using clinical data collected from four hospitals. The validation results demonstrated the model's robust generalizability across diverse clinical settings.

Third, on both the external and the internal test sets, the Swin Transformer demonstrated superior prediction accuracy compared with two intermediate-level clinicians and two junior-level clinicians. Furthermore, we combined expert clinician predictions with those from the deep learning model. This integration led to improvements in sensitivity, specificity, AUC, accuracy, and diagnostic consistency among clinicians. Based on the findings of this research, The AI model proposed in this study serves as an auxiliary decision-support tool, providing clinicians with an objective assessment based on the likelihood of spontaneous passage of radiologically detectable urinary stones. This tool is particularly designed to assist junior and mid-level clinicians in clinical decision-making.

Despite these promising results, our research has certain limitations. First, this was a multicentre retrospective study, and no prospective cohort validation was performed. Performing additional prospective studies involving larger sample sizes and multiple centres would validate our results. Second, we utilized only two characteristic CT slices per patient to visualize ureteral stones. The prediction model learns directly from the original or preprocessed images and has no intuitive clinical parameters. Future work will focus on incorporating information from all available slices for a more comprehensive analysis. Finally, The lack of a washout period in the evaluation of model-assisted versus unaided clinical decision-making may have introduced recall bias, potentially affecting the reliability of performance assessments. our model currently lacks integration with other relevant clinical data, such as pain scores, inflammatory markers, and stone location. The incorporation of such data could increase the accuracy of the model.

Conclusions

This study highlights the viability and promises of using a Swin Transformer-based deep learning model for predicting the spontaneous passage of ureteral stones. This model, trained and validated on a multicentre dataset, shows promise for improving diagnostic accuracy and assisting clinicians in managing patients with ureteral calculi. Future studies will focus on addressing the limitations of our study, conducting prospective and real-world validations, and improving the model's performance and clinical applicability.

Appendix

See Table 4.

Table 4 Comparison of indicators across four external entities

	DL	RU	DL+RU	AU	DL+AU	SU	DL+SU	RR	DL+RR	AR	DL+AR	SR	DL+SR
Auc													
DL		<0.0001	<0.0001	<0.0001	<0.0001	0.0029	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RU			<0.0001	0.02655728	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RU				<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AU					<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+AU						<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.06063	<0.0001	<0.0001
SU							<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+SU								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RR									<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AR										<0.0001	<0.0001	<0.0001	0.3568
DL+AR											<0.0001	<0.0001	<0.0001
SR												<0.0001	<0.0001
DL+SR													<0.0001
AUC													
DL		<0.0001	<0.0001	<0.0001	<0.0001	0.001977	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RU			<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.000248	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RU				<0.0001	0.031728	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.112667	<0.0001	<0.0001
AU					<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+AU						<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.534491	<0.0001	<0.0001
SU							<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+SU								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RR									<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AR										<0.0001	<0.0001	<0.0001	0.219082
DL+AR											<0.0001	0.065721	<0.0001
SR												<0.0001	<0.0001
DL+SR													<0.0001
F1													
DL		<0.0001	0.832234	<0.0001	<0.0001	0.007609	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RU			<0.0001	0.002407	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RU				<0.0001	<0.0001	0.005527	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AU					<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.007528	<0.0001
DL+AU						<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.003127	<0.0001	<0.0001
SU							0.001055	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+SU								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RR									<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AR								0.002577			<0.0001	<0.0001	0.617804
DL+AR											<0.0001	<0.0001	0.005143
												<0.0001	<0.0001

Table 4 (continued)

	DL	RU	DL+RU	AU	DL+AU	SU	DL+SU	RR	DL+RR	AR	DL+AR	SR	DL+SR
SR													<0.0001
DL+SR													
Sen													
DL	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.66747	0.009887	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RU	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RU	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AU	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+AU	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.091141	<0.0001	<0.0001	<0.0001
SU						0.003573	0.003573	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+SU								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RR								<0.0001	<0.0001	<0.0001	<0.0001	0.01584	<0.0001
AR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.604847
DL+AR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
SR												<0.0001	<0.0001
DL+SR												<0.0001	<0.0001
Spe													
DL	<0.0001	<0.0001	<0.0001	<0.0001	0.054044	0.000203	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.12452	<0.0001
RU	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RU	<0.0001	<0.0001	<0.0001	0.000437	<0.0001	<0.0001	<0.0001	0.014564	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
AU					<0.0001	<0.0001	<0.0001	0.18889	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+AU					<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.632564	0.014077
SU						0.002761	0.002761	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+SU								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.041434
AR								<0.0001	<0.0001	<0.0001	0.714719	<0.0001	<0.0001
DL+AR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
SR											<0.0001	<0.0001	0.101693
DL+SR											<0.0001	<0.0001	0.002449
Pre													
DL	<0.0001	<0.0001	<0.0001	<0.0001	0.000653	0.000225	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RU	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.40191	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RU	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.001189	<0.0001	<0.0001	<0.0001	0.083965
AU					<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.106027	<0.0001	<0.0001	<0.0001
DL+AU					<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.092264	<0.0001	<0.0001
SU						0.037012	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+SU								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001
DL+RR								<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.128361

Table 4 (continued)

	DL	RU	DL+RU	AU	DL+AU	SU	DL+SU	RR	DL+RR	AR	DL+AR	SR	DL+SR
AR											<0.0001	<0.0001	<0.0001
DL+AR											<0.0001	<0.0001	<0.0001
SR												<0.0001	0.000331
DL+SR													

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval This study was conducted according to the guidelines of the Declaration of Helsinki, and approved by Ethics Committee of Ningde Hospital Affiliated to Ningde Normal University (NDYKYL-2024-34).

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